Arunabha Majumder

Lab Engineer | Experimental Setups, Sensors, Calibration and Data Acquisition

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Department of Mechanical and Production Engineering, Workshop and Laboratories
Aarhus University, Katrinebjergvej 89C, 8200 Aarhus N

Application: Lab Engineer in Mechanical Engineering (ID 170084)

Dear Peter Lykke and the hiring committee,

I am applying for the Lab Engineer position in your workshop and laboratory unit. I hold an MTech in Mechatronics and a BTech in Mechanical Engineering, and I am finishing my PhD at Aalborg University, where most of my days are spent in the lab. Reading the ad, I recognised my own work: building setups, fitting sensors, calibrating them and helping other people get good data.

On a 5-bar parallel robot I calibrated a force estimate from variable-stiffness actuators against a reference force sensor and got it within 10% RMS error. We then used that setup to measure human arm impedance in the gait lab. For my exoskeleton project I built and maintained the test rigs, with a motor, encoder, force sensors and DAQ, and I collected data from 12 participants using EMG, Qualisys motion capture and force/torque sensors. That work meant writing protocols, keeping recordings in order and thinking about participant safety at every session.

Before Aalborg I spent two years at CSIR-CMERI, a national research lab in India. There I fabricated and assembled pneumatic artificial muscles to an existing design, collected data on the lab's setup and worked on the PID part of a gain-scheduled controller in MATLAB/Simulink. My tools are MATLAB, Python, LabVIEW at a basic level and Arduino, and I design fixtures in SolidWorks and print or machine them as needed.

I also like the teaching side of a lab. I have taught university students and supervised Bachelor's students since 2023, often helping them set up equipment and understand strange readings. I hold a Diploma in Fine Arts, and I use it to make clear sketches and figures for lab guides and reports, which helps when a student needs to rebuild a setup alone.

I have less hands-on experience with acceleration, temperature and flow sensors than with force and pressure, and I would be glad to learn them from your team. I expect to defend my PhD in 2027 and would like to discuss a start date that works for the department.

Thank you for considering my application. I would be happy to visit Katrinebjergvej and talk about the role.

Kind regards,
Arunabha Majumder